Cybersecurity Mini Project – Secure Network Design and Implementation Using Cisco Packet Tracer

1. Introduction

This project focused on designing and implementing a secure network topology for a small office environment consisting of approximately 25 employees. The implementation was completed using Cisco Packet Tracer and applied core networking and cybersecurity principles including VLAN segmentation, inter-VLAN routing, access control, and secure network design.

The objective was to provide reliable connectivity while protecting organizational resources through logical network segmentation and controlled communication between different departments.

2. Network Requirements

The office network was designed to support the following resources:

- Five employee workstations
- One file server
- Two network printers
- One management laptop
- Internet connectivity (represented by the ISP cloud/router)
- Secure communication between network segments
- Administrative management network
- Guest network isolation

The design follows defense-in-depth principles by separating users, servers, guests, and management devices into different VLANs.

3. Network Topology

The implemented topology consists of:

- ISP Cloud
- Cisco Router
- Cisco 3560 Multilayer Switch
- Cisco 2960 Access Switch
- Five Employee PCs

- One Server
- Two Printers
- One Management Laptop

Traffic flows through the ISP cloud to the router, then to the multilayer switch. The multilayer switch performs inter-VLAN routing and connects to the access switch using an IEEE 802.1Q trunk link. End devices connect to the access switch according to their assigned VLAN.

Insert Screenshot 1: Complete Network Topology

4. VLAN Segmentation Strategy

To improve security and reduce unnecessary network traffic, the network was divided into four VLANs.

VLAN	Name	Purpose
10	Employees	Employee workstations
20	Servers	File server and printers
30	Guest	Guest wireless devices
40	Management	Administrator laptop

This segmentation prevents all devices from sharing the same broadcast domain and improves security by separating sensitive systems from general user traffic.

Insert Screenshot 2: show vlan brief

5. IP Addressing Plan

Each VLAN was assigned its own subnet.

VLAN	Network	Default Gateway
VLAN 10	192.168.10.0/24	192.168.10.1
VLAN 20	192.168.20.0/24	192.168.20.1
VLAN 30	192.168.30.0/24	192.168.30.1
VLAN 40	192.168.40.0/24	192.168.40.1

Device Address Assignment

Device	IP Address
PC1	192.168.10.10
PC2	192.168.10.11
PC3	192.168.10.12

PC4	192.168.10.13
PC5	192.168.10.14
File Server	192.168.20.10
Printer 1	192.168.20.20
Printer 2	192.168.20.21
Management Laptop	192.168.40.10

The Cisco 3560 multilayer switch was configured with Switch Virtual Interfaces (SVIs) to provide gateway services for each VLAN.

Insert Screenshot 3: show ip interface brief

6. Trunk Configuration

A trunk connection was configured between the Cisco 3560 multilayer switch and the Cisco 2960 access switch using GigabitEthernet0/1.

The trunk allows multiple VLANs to traverse a single physical connection using IEEE 802.1Q tagging.

This design reduces cabling requirements while maintaining logical separation of VLAN traffic.

Insert Screenshot 4: show interfaces trunk

7. Security Controls

Several security mechanisms were incorporated into the design.

VLAN Segmentation

Employees, servers, guests, and management devices were isolated into separate VLANs to reduce the attack surface.

Inter-VLAN Routing

Routing between VLANs was performed on the Cisco 3560 multilayer switch using Switch Virtual Interfaces (SVIs). The ip routing command was enabled to allow communication between authorized VLANs.

Access Control List (ACL)

An extended ACL (ACL 100) was created to restrict Guest VLAN traffic.

The ACL blocks:

- Guest VLAN → Employee VLAN
- Guest VLAN → Server VLAN

The final rule permits all other traffic.

ACL Configuration:

- Deny traffic from 192.168.30.0/24 to 192.168.10.0/24
- Deny traffic from 192.168.30.0/24 to 192.168.20.0/24
- Permit all remaining traffic

This implementation demonstrates role-based network access control by limiting guest users from accessing sensitive internal resources.

Insert Screenshot 5: show access-lists

8. Connectivity Testing

After configuration, connectivity testing was performed using the ping utility.

A successful ping was completed from an employee workstation (PC1) to the file server located in a different VLAN.

This confirmed that:

- VLAN configuration was correct.
- Trunk links were operational.
- Inter-VLAN routing was functioning.
- IP addressing was correctly configured.

Insert Screenshot 6: Successful Ping Test

9. Monitoring and Security Recommendations

To maintain the security and performance of the network, the following monitoring strategy is recommended:

- Centralized logging using a Syslog server.
- Packet analysis using Wireshark.
- Regular vulnerability assessments.
- Continuous monitoring of switch and router logs.
- Periodic review of ACLs and firewall policies.
- Configuration backups after network changes.

These practices improve visibility into network activity and support early detection of security incidents.

10. Risk Mitigation

The implemented design reduces several common cybersecurity risks.

Threat	Mitigation
Unauthorized internal access	VLAN segmentation
Lateral movement	Access Control Lists
Broadcast storms	Separate broadcast domains
Unauthorized guest access	Guest VLAN isolation
Network misconfiguration	Structured IP addressing
Unauthorized administrative access	Dedicated Management VLAN

The layered approach follows the principle of defense-in-depth by combining network segmentation, access control, and logical isolation.

11. Conclusion

This project successfully demonstrated the implementation of a secure small office network using Cisco Packet Tracer. The completed solution incorporated VLAN segmentation, inter-VLAN routing, trunk configuration, structured IP addressing, and access control through extended ACLs.

The design improves confidentiality, integrity, and availability by isolating network resources and controlling communication between different network segments. The project also reflects industry best practices aligned with defense-in-depth principles and provides a scalable foundation for future network expansion.